



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



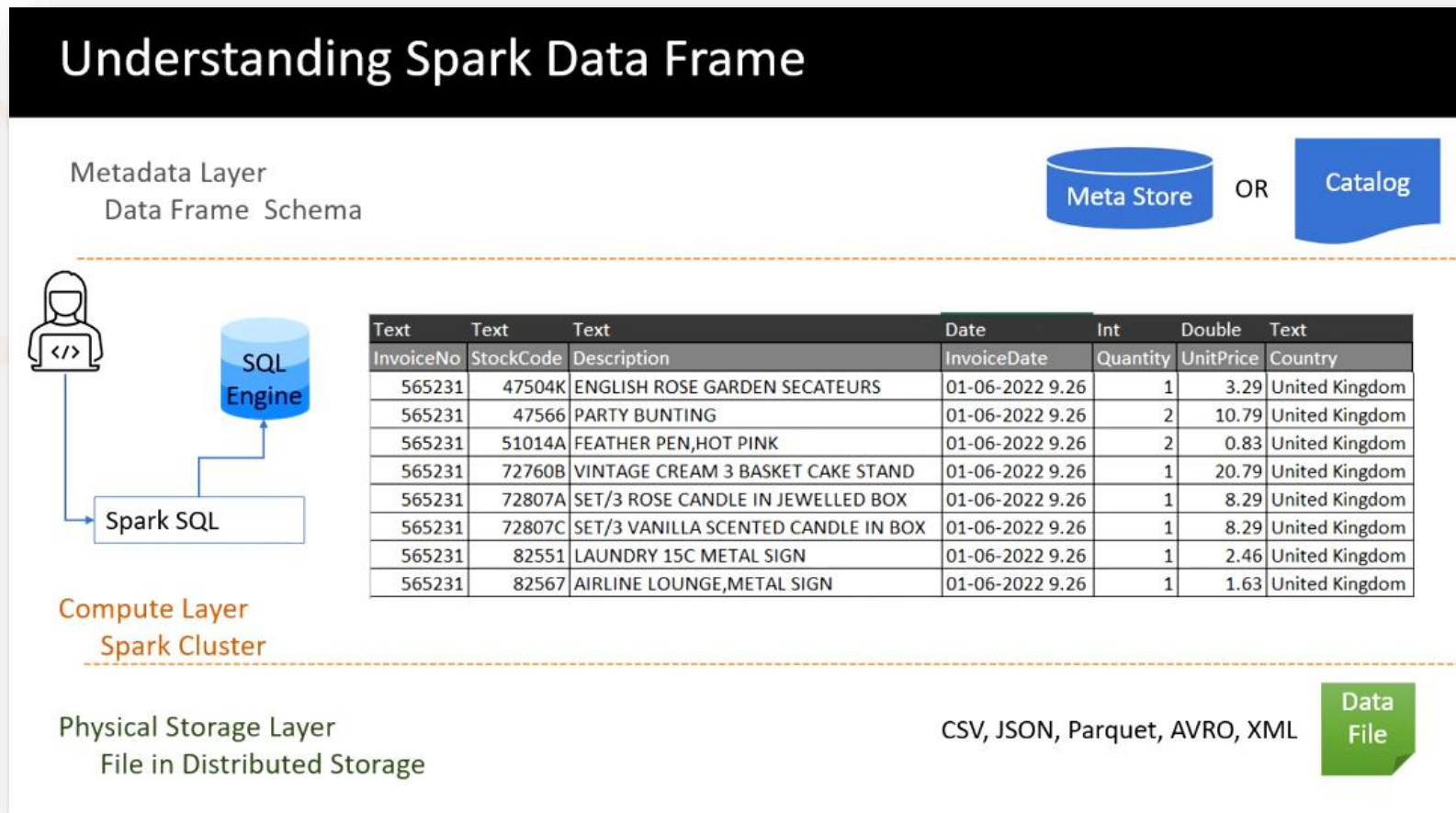
Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Problems with Databricks Community

Before we proceed further with the micro project, let us have a look at the below diagram once again. Every Spark Table or the Spark Dataframe lives in three layers:

1. Layer – It stores Table or Dataframe definition and the schema information.
2. Compute Layer – It runs your Spark SQL engine.
3. Physical Storage Layer – It stores the spark data in a data file.



When you execute a Spark SQL query, or you run a Dataframe code, the Spark SQL engine will refer to the metadata store. And the metadata will tell the following things:

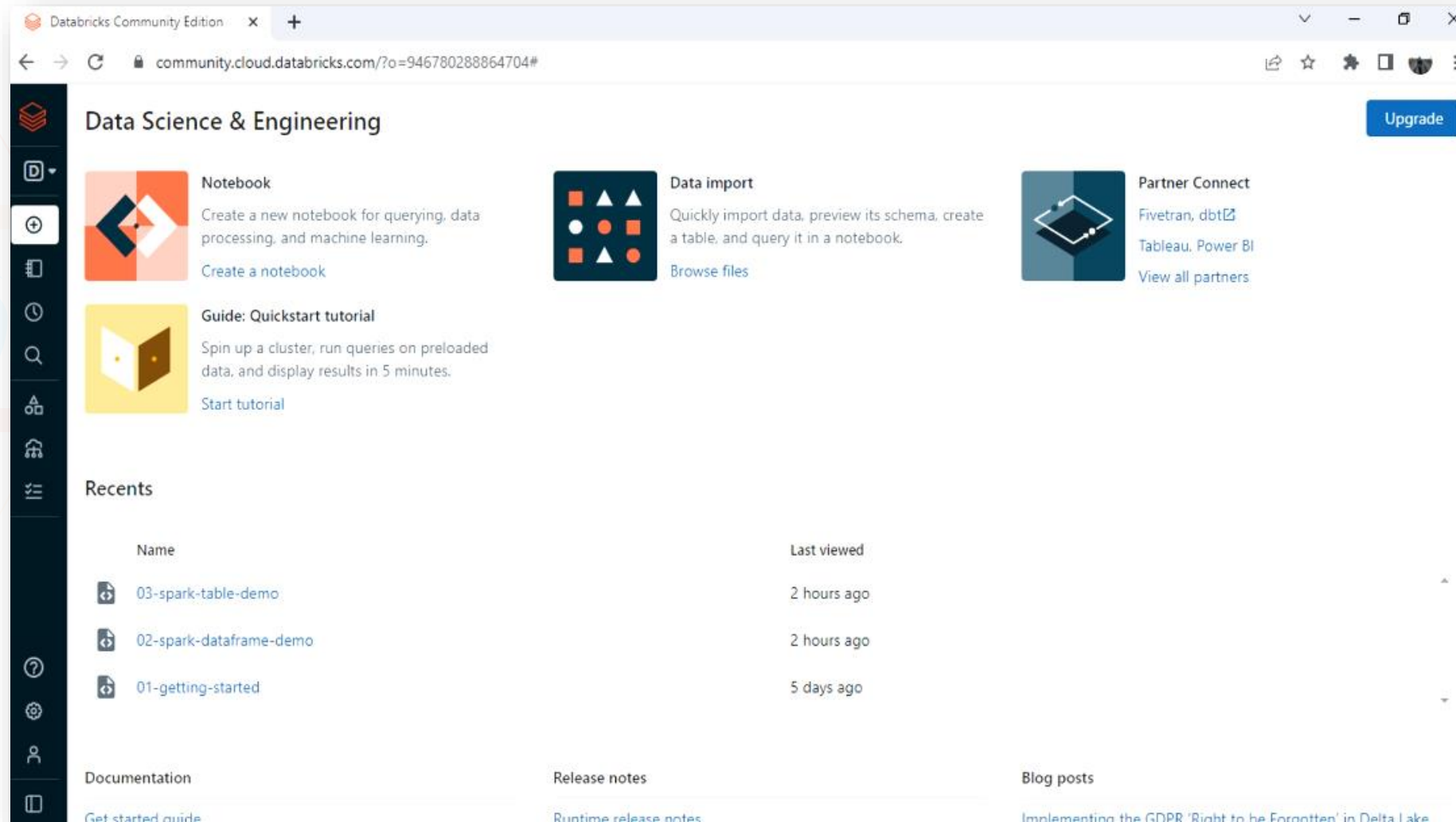
1. Where is the data for this table?
2. What is the schema of the data and table?

If the Spark SQL engine doesn't find metadata data for the table, it throws an analysis exception.

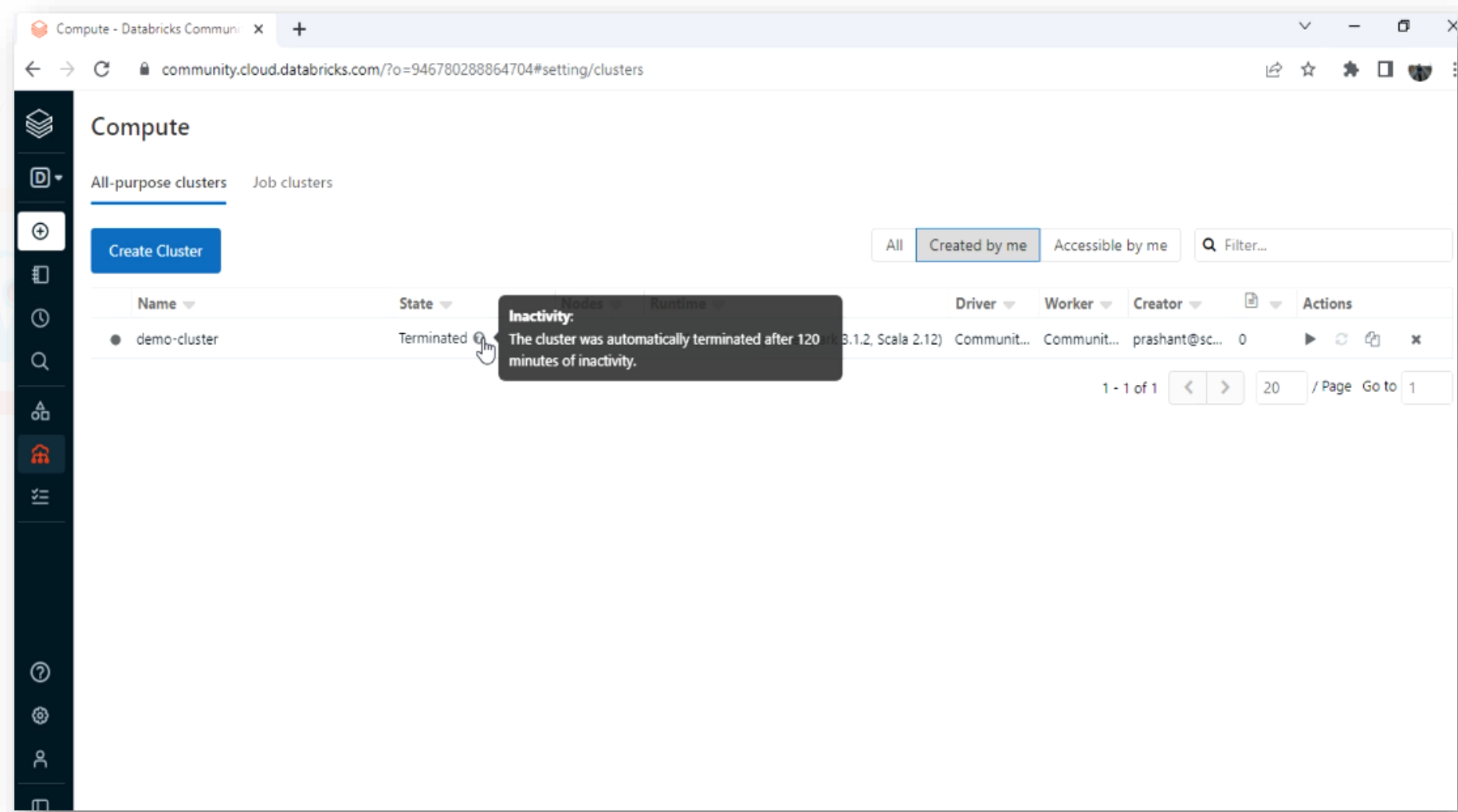
If metadata is there, the SQL engine will read the data from the data file and present it to you like a table or Dataframe. So, the three(Metadata, Compute and Storage) layers of Spark are supercritical.

You will face problems and see errors when any of these three layers are missing or broken.

Go to your Databricks workspace and go to the compute menu.



You need a running cluster for doing anything in the Databricks environment because you need a compute layer. But we have an old cluster here which is in terminated state because I left it idle for more than 2 hours, so Databricks terminated my cluster.



I cannot restart this cluster again. I must create a new cluster if I want to work again. Databricks community edition will terminate your cluster after 2 hours of idle time, and once your cluster is terminated, Databricks will clean up the following things:

- 1. Cluster VM (Compute Layer) – You cannot restart your cluster again.
- 2. Spark Metadata Store (Metadata Layer) – You will lose your databases and tables.

Compute

All-purpose clusters

Job clusters

Create Cluster

All

Created by

Could not start cluster demo-cluster.
Cluster Start feature is currently disabled.

Name	State	Nodes	Runtime	Driver	Worker	Creator		Actions
demo-cluster	Terminated	-	9.1 LTS (includes Apache Spark 3.1.2, Scala 2.12)	Communit...	Communit...	prashant@sc...	0	...

1 - 1 of 1

<

>

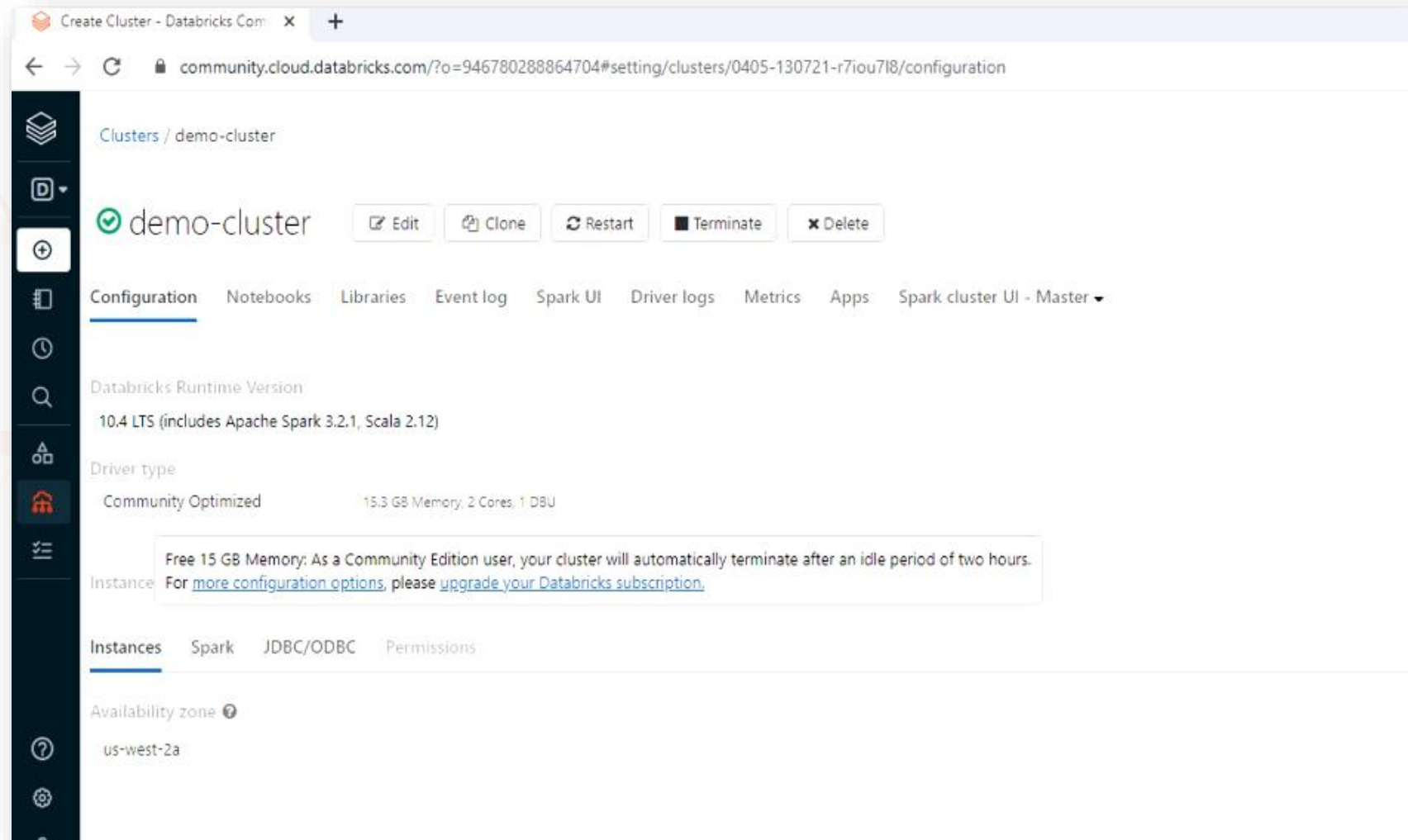
20

/ Page

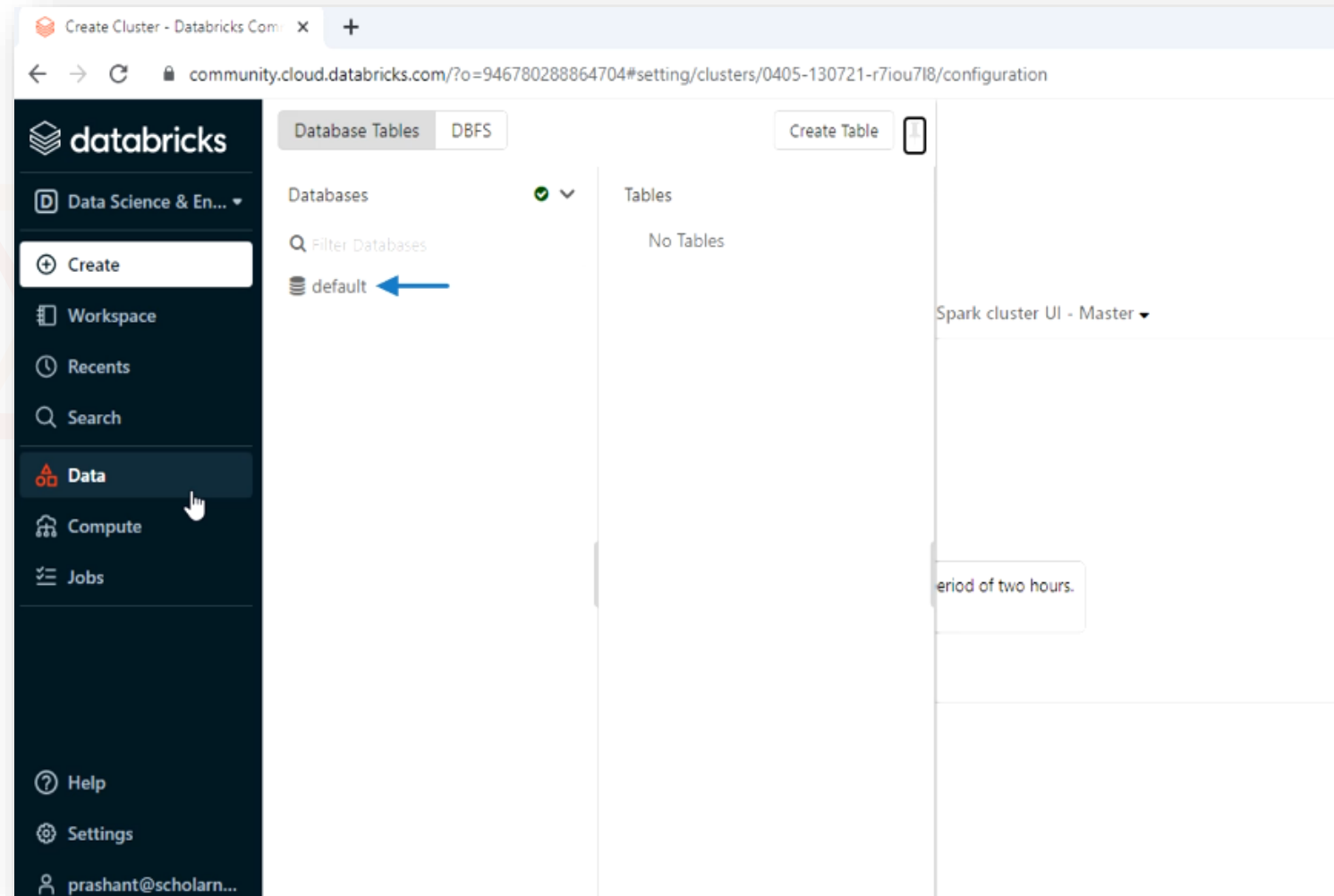
Go to

1

You can delete the old cluster, and create a new one with same name. Creating a cluster takes about 5-10 minutes.

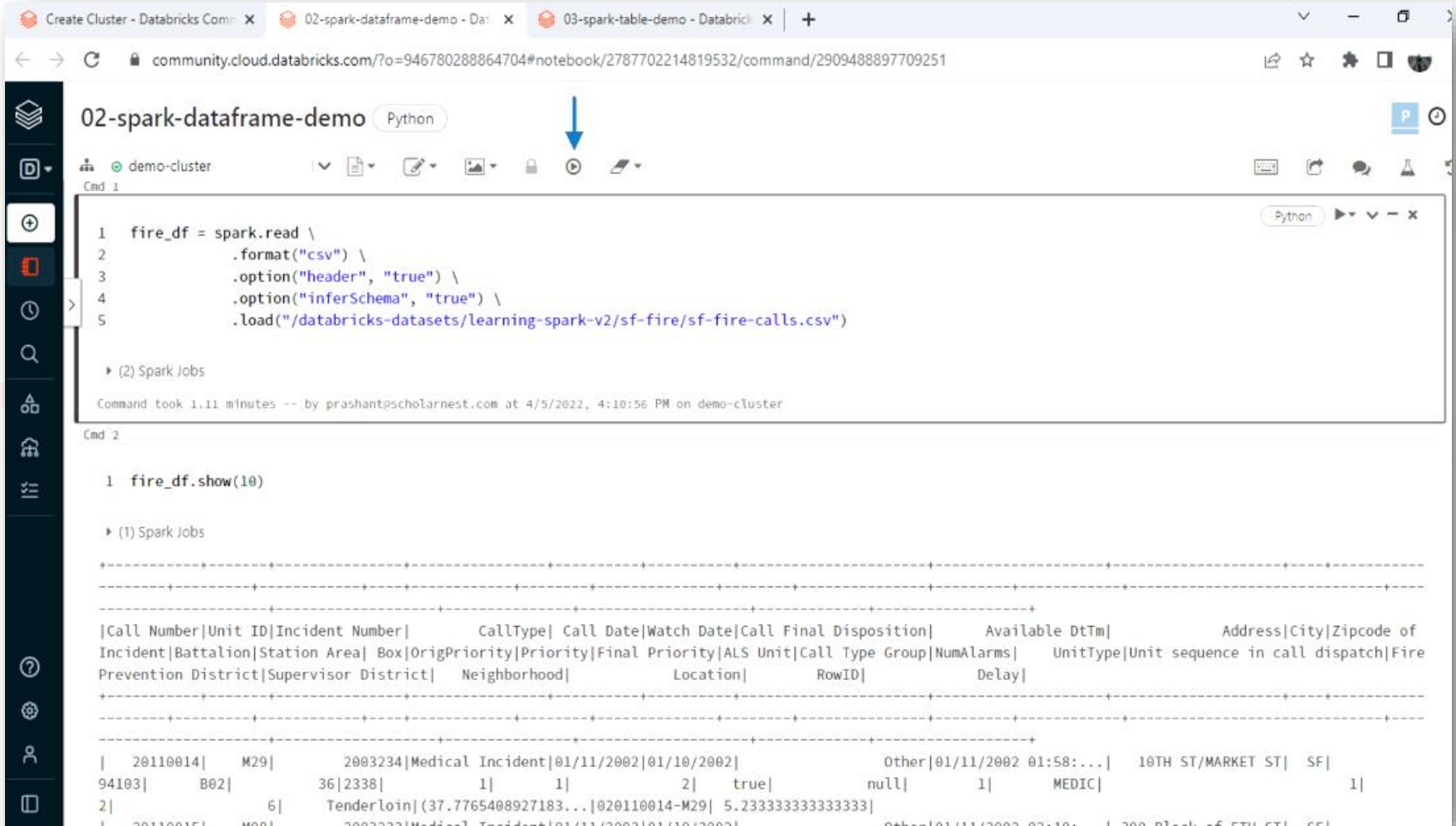


You can click the data menu, and you can see the database and tables. We only have a default database the *demo_db* database that we created yesterday is gone. Because Databricks cleaned my Spark metadata store as my cluster was idle for more than 2 hours. So I lost my database and table definition. This problem does not exist for the full licensed version of the Databricks Cloud.



The community edition will clean up the VM and the metadata store. But it doesn't clean up the directories and files. So you do not lose your storage layer. Your data files remain in place even if your compute and metadata layer is gone. Directories are created in an external distributed storage, and files are stored in these directories on external storage.

You can go the notebooks you created earlier (02-spark-dataframe-demo.ipynb and 03-spark-table-demo.sql) from the workspace menu option, and click the run all option shown below to run the entire notebook at once. Make sure your cluster is attached to both the notebooks.



02-spark-dataframe-demo Python

demo-cluster

Cmd 1

```
1 fire_df = spark.read \  
2     .format("csv") \  
3     .option("header", "true") \  
4     .option("inferSchema", "true") \  
5     .load("/databricks-datasets/learning-spark-v2/sf-fire/sf-fire-calls.csv")
```

▶ (2) Spark Jobs

Command took 1.11 minutes -- by prashant@scholarnest.com at 4/5/2022, 4:10:56 PM on demo-cluster

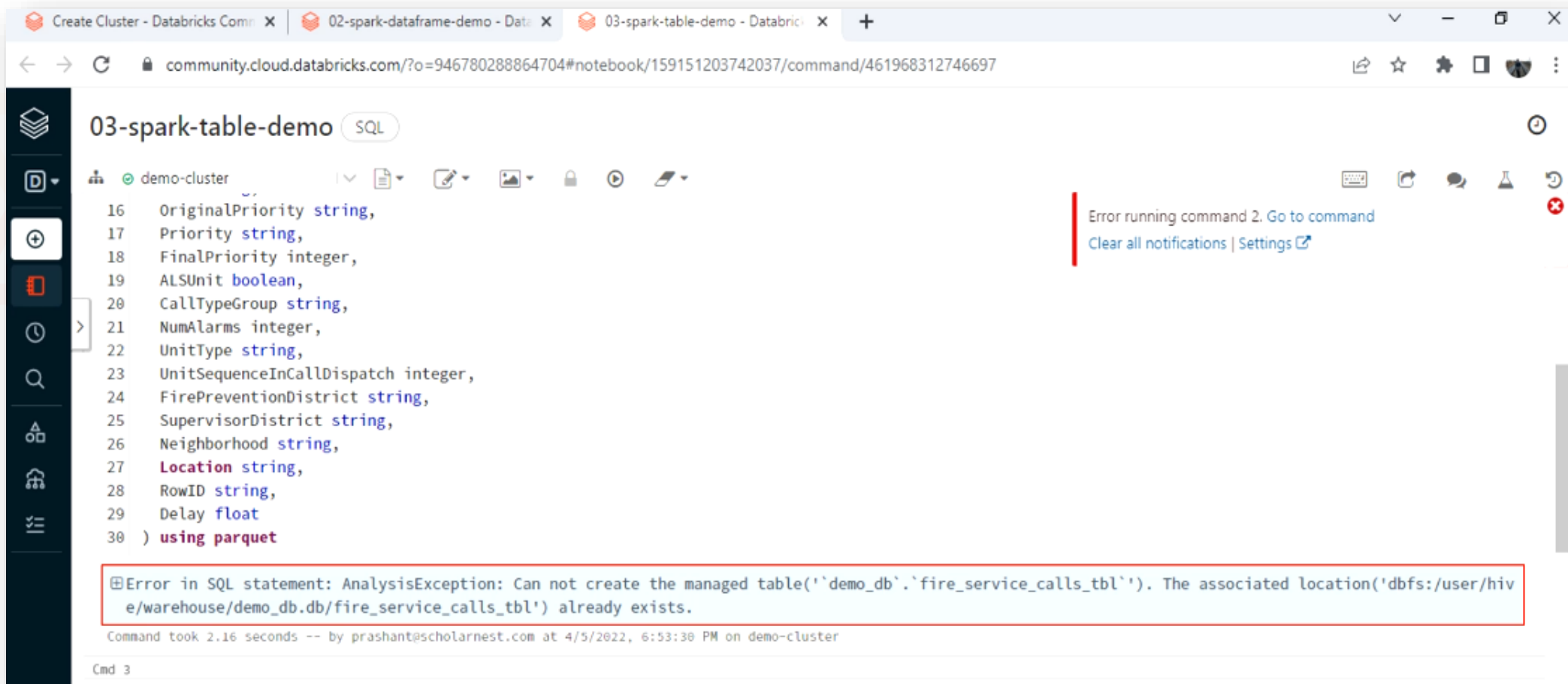
Cmd 2

```
1 fire_df.show(10)
```

▶ (1) Spark Jobs

Call Number	Unit ID	Incident Number	CallType	Call Date	Watch Date	Call Final Disposition	Available DtTm	Address	City	Zipcode of
94103	B02	36 2338	1	1	2	true	null	1	MEDIC	1
2	6	Tenderloin	(37.7765408927183...	020110014-M29	5.233333333333333					

We ran the *02-spark-dataframe-demo.ipynb* successfully but encountered an error in the *03-spark-table-demo.sql* notebook. The error says that “can not create the managed table. The associated location already exists.” We didn't face any errors yesterday when we created this table. But we are getting an error today because Databricks didn't clean the directory and data files. My storage layer is still there. So my data directory is not cleaned.



```
16 OriginalPriority string,
17 Priority string,
18 FinalPriority integer,
19 ALSUnit boolean,
20 CallTypeGroup string,
21 NumAlarms integer,
22 UnitType string,
23 UnitSequenceInCallDispatch integer,
24 FirePreventionDistrict string,
25 SupervisorDistrict string,
26 Neighborhood string,
27 Location string,
28 RowID string,
29 Delay float
30 ) using parquet
```

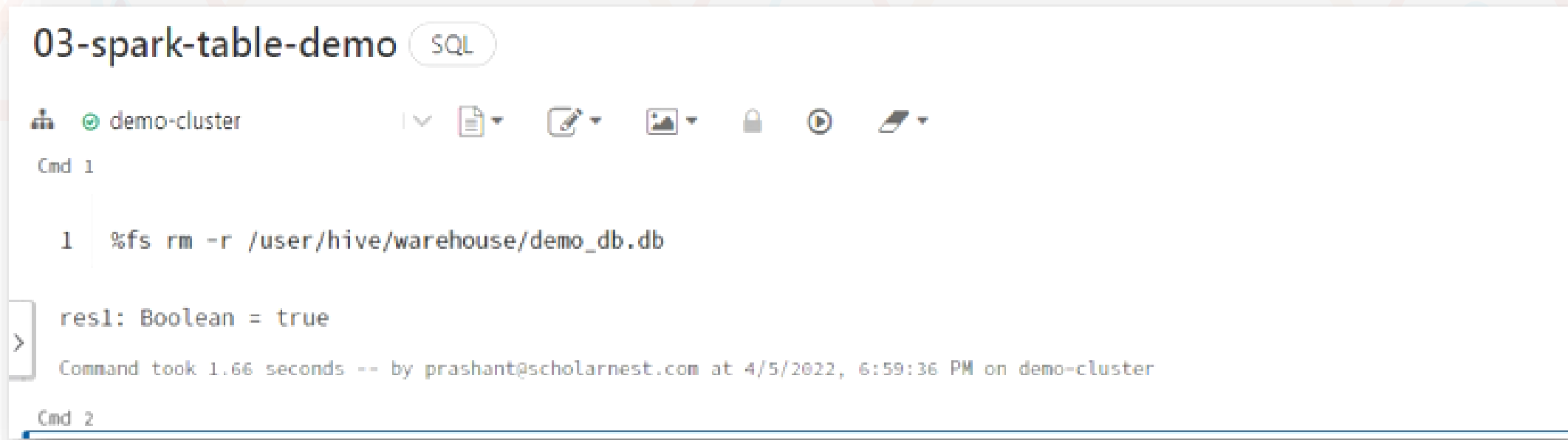
Error running command 2. Go to command
Clear all notifications | Settings

Error in SQL statement: AnalysisException: Can not create the managed table('demo_db`.`fire_service_calls_tbl`). The associated location('dbfs:/user/hive/warehouse/demo_db.db/fire_service_calls_tbl') already exists.

Command took 2.16 seconds -- by prashant@scholarnest.com at 4/5/2022, 6:53:30 PM on demo-cluster

Cmd 3

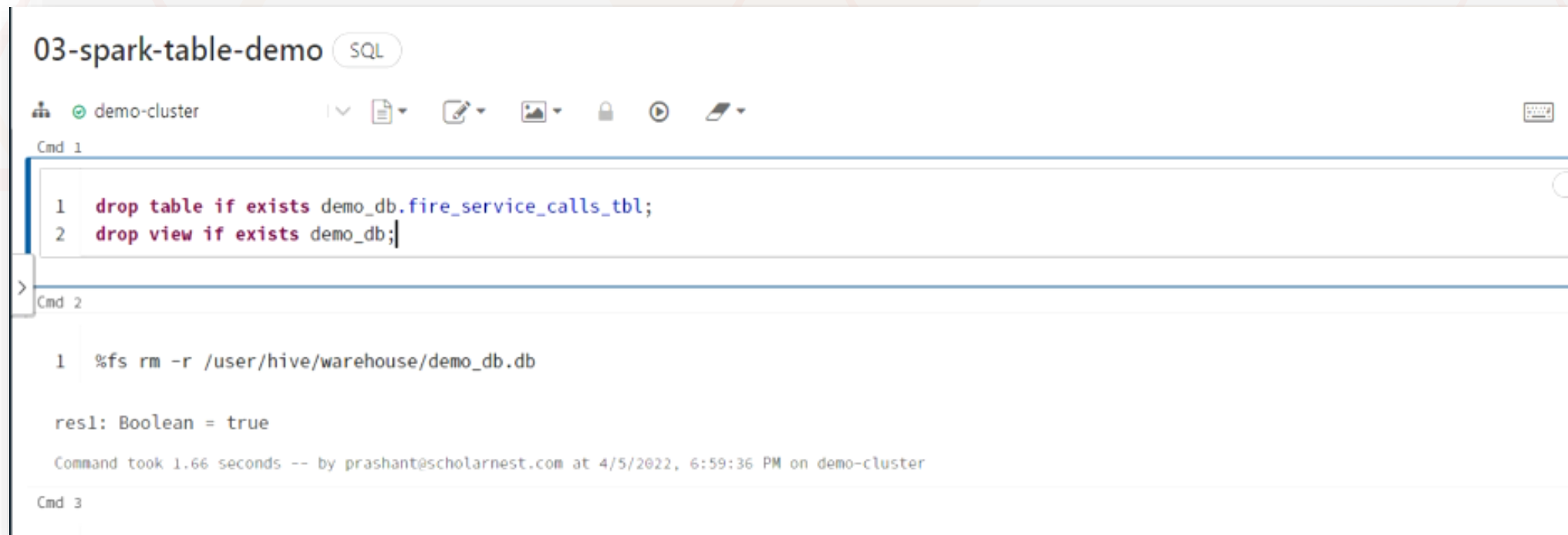
In order to fix the Analysis Exception, we can add a new cell at the top of the notebook and type the filesystem rm command to clean the directory. Copy the directory location from the error message.



The screenshot shows a Databricks notebook titled "03-spark-table-demo" with a "SQL" language selector. The notebook is running on a "demo-cluster". A command cell labeled "Cmd 1" contains the command `%fs rm -r /user/hive/warehouse/demo_db.db`. The output of the command is `res1: Boolean = true`. Below the output, a status message reads: "Command took 1.66 seconds -- by prashant@scholarnest.com at 4/5/2022, 6:59:36 PM on demo-cluster". A second command cell labeled "Cmd 2" is visible at the bottom of the notebook.

```
03-spark-table-demo SQL
demo-cluster
Cmd 1
1 %fs rm -r /user/hive/warehouse/demo_db.db
res1: Boolean = true
Command took 1.66 seconds -- by prashant@scholarnest.com at 4/5/2022, 6:59:36 PM on demo-cluster
Cmd 2
```

I also recommend adding a drop table and drop database command at the top. You can add another cell at the top of the notebook and run two DDL statements to drop your table and database. Do not forget to add a semicolon at the end of the SQL expression.



03-spark-table-demo SQL

demo-cluster

Cmd 1

```
1 drop table if exists demo_db.fire_service_calls_tbl;  
2 drop view if exists demo_db;
```

>

Cmd 2

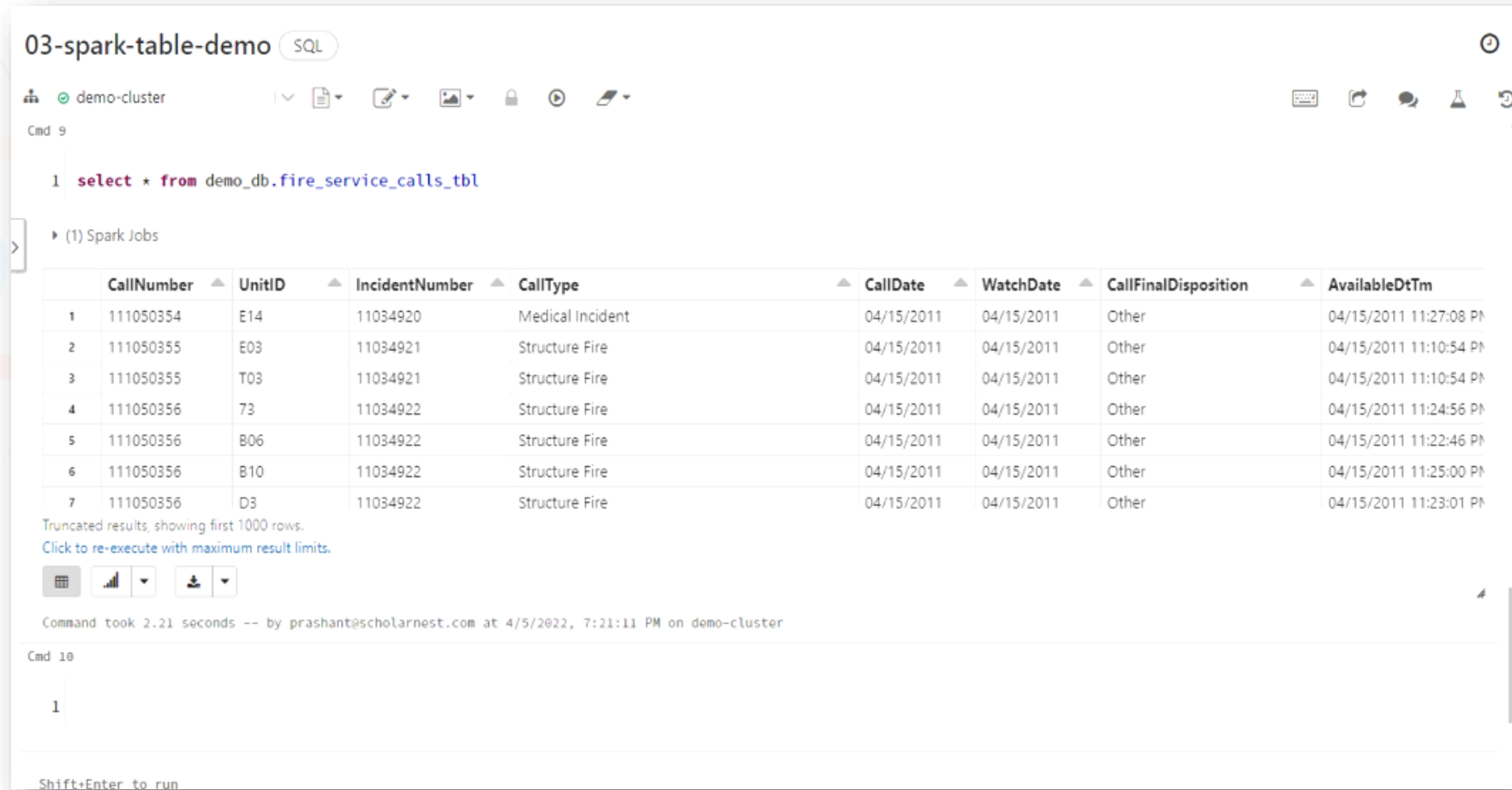
```
1 %fs rm -r /user/hive/warehouse/demo_db.db
```

res1: Boolean = true

Command took 1.66 seconds -- by prashant@scholarnest.com at 4/5/2022, 6:59:36 PM on demo-cluster

Cmd 3

Now, you can go ahead and click the run all button for the *03-spark-table-demo.sql* notebook and it should work perfectly fine without giving any errors.



03-spark-table-demo SQL

demo-cluster

Cmd 9

```
1 select * from demo_db.fire_service_calls_tbl
```

(1) Spark Jobs

	CallNumber	UnitID	IncidentNumber	CallType	CallDate	WatchDate	CallFinalDisposition	AvailableDtTm
1	111050354	E14	11034920	Medical Incident	04/15/2011	04/15/2011	Other	04/15/2011 11:27:08 PM
2	111050355	E03	11034921	Structure Fire	04/15/2011	04/15/2011	Other	04/15/2011 11:10:54 PM
3	111050355	T03	11034921	Structure Fire	04/15/2011	04/15/2011	Other	04/15/2011 11:10:54 PM
4	111050356	73	11034922	Structure Fire	04/15/2011	04/15/2011	Other	04/15/2011 11:24:56 PM
5	111050356	B06	11034922	Structure Fire	04/15/2011	04/15/2011	Other	04/15/2011 11:22:46 PM
6	111050356	B10	11034922	Structure Fire	04/15/2011	04/15/2011	Other	04/15/2011 11:25:00 PM
7	111050356	D3	11034922	Structure Fire	04/15/2011	04/15/2011	Other	04/15/2011 11:23:01 PM

Truncated results, showing first 1000 rows.
[Click to re-execute with maximum result limits.](#)

Command took 2.21 seconds -- by prashant@scholarnest.com at 4/5/2022, 7:21:11 PM on demo-cluster

Cmd 10

```
1
```

Shift+Enter to run



Thank You
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